Nancy Ihechu

Email:nancyihechu@gmail.com || Phone: +1 573 814 8366|| LinkedIn ||GitHub

SUMMARY

Computational chemist with experience developing machine learning methods for drug discovery. Skilled in free energy calculations, and molecular modeling. Passionate about developing computational methods that accelerate scientific discovery.

EDUCATION

Ph.D. Chemistry (Computational Drug Discovery) **Jan.2026 – Present**
University of Missouri, USA

M.Sc. Chemistry/Chemo-Informatics **Sep.2023 – July.2025**
University of Strasbourg, France (AI in Chemistry)
University of Ljubljana, Slovenia (Chemistry)

B.Sc. Chemistry **Jan.2015 – Feb.2019**
Imo State University, Nigeria

WORK EXPERIENCE

Graduate Research Assistant **May 2025 – Present**
University of Missouri-Columbia, USA

- Secured a competitive $5,000 research grant to develop a quantum machine learning model for classifying and prioritizing potential drug candidates for LRRK2 PROTAC
- Developed and benchmarked ensemble machine learning models using a DNA-encoded library (DEL) of over 7.7 million compounds to identify experimentally confirmed PGK2 actives for a male contraceptive drug discovery project
- Evaluated the binding thermodynamics of over 100 ligand–protein complexes using Absolute Binding Free Energy calculations in GROMACS, supporting hit candidate prioritization for Parkinson's disease drug discovery
- Applied similarity-based virtual screening to the Enamine REAL library of over 100 billion compounds, identifying LRRK2 candidate molecules for downstream experimental validation
- Reduced molecular dynamics simulation time by 60% through GPU-optimized HPC pipelines, Slurm job scheduling, and large-scale parallel execution

Summer Intern **March.2025 – April.2025**
Novartis Slovenia

- Developed automated Python workflows for Host Cell Protein (HCP) data extraction, transformation, visualization, and reporting, contributing to a project aimed at reducing biological data processing time
- Improved accessibility of HCP datasets by developing interactive dashboards for real-time monitoring of protein data and project progress, supporting cross-functional research teams
- Communicated analytical insights by presenting clear visualization of data-driven findings that supported project planning and operational efficiency

Laboratory Analyst **Dec.2019 – July.2023**
Standards Organization of Nigeria

- Performed chromatography and mass spectrometry analyses on environmental samples, supporting analytical data generation, quality assessment, and regulatory testing
- Improved interpretation of analytical results by developing Python- and Power BI-based workflows that automated quality-control reporting and supported a team of 10 laboratory scientists
- Maintained regulatory compliance and scientific documentation by preparing quality-control records, interpreting analytical results, and ensuring accurate laboratory record-keeping
- Managed laboratory inventories, coordinating equipment maintenance schedules, and ensuring the availability of research supplies

PUBLICATION

Ihechu, N. I. Anti-Parkinson's Therapeutic Discovery: A Computational Approach. Master's Thesis, University of Ljubljana, Faculty of Chemistry and Chemical Technology, 2026 PID: 20.500.12556/RUL-182772

PRESENTATIONS

American Chemical Society Regional Meeting **Oct. 2025**

- Title: Anti-Parkinson's Therapeutic Discovery: A Computational Approach
- Description: Poster Presentation on prioritization workflow in the early phase drug discovery

HONORS AND AWARDS

Erasmus Mundus Double Master's Degree Scholarship **Sep. 2023**

- Description: Competitive European Union-funded scholarship awarded for a joint international master's program in Chemistry and Chemoinformatics.

AFFLIATION

- American Chemical Society (ACS) – Member
- Machine Learning Innovation Network For Research to Advance MEdicinal Chemistry (MAINFRAME) – Member